Type MS · Explain how villi aid absorption and the effect of damage

Answer 1 of 7 · Question 5 part c.ii · 3 marks

2025 JUN · Paper 02 (regional R) · find part c.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Explain how villi aid absorption and the effect of damage

Answer 2 of 7 · Question 4 part c.iii · 2 marks

2020 JAN · Paper 02 · find part c.iii in this mark scheme extract

Number				
4	(a)(i)	pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1)	Accept fatigue	1
	(ii)	Any two from: • itching transfers eggs (from anus) to fingers/hand(1) • eggs transferred from fingers to other surfaces/people(1) • eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1)		2
	(b)	Any two from: • correct temperature/pH/provides warmth(1) • plenty of nutrients(1) • no predators(1)	Allow food	2
	(c) (i)	lacking in one or more nutrients/vitamins/minerals / other named nutrient(1)		1
	(ii)	any named nutritional deficiency such as anaemia/scurvy	Allow named nutrient e.g. protein deficiency	1
	(iii)	• pinworms absorb nutrients/named nutrient(1) • less nutrients absorbed/pass into blood/body/cells receive less nutrients(1)	Allow food	2
	(d)	Reference to eggs/pinworms on transparent tape (where they can be seen)		1
	(e)	Any two from: • refrain from itching/scratching (anus)(1) • use medication (to kill pinworms)(1) • don't share towels/clothing(1) • wash bedclothes/linen regularly/do not shake bed linen(1) • don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1) • don't allow children to bathe together(1)		2

Total for Question 4 = 12 marks

Type MS · Explain how villi aid absorption and the effect of damage

Answer 3 of 7 · Question 4 part b · 4 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

4	(a)	(i)	line labelled X to mouth;		1mark
		(ii)	line labelled Y to stomach;		1mark
		(iii)	line labelled Z to pancreas;		1mark
	(b)		any 4 of • reduced surface area; • so unable to absorb calcium; • required for bones/compact bone; • calcium passes from bone; • causes them to weaken/soften/lose density;		4marks
	(c)		• large intestine absorbs water; • not enough absorbed;		2marks

Total 9 marks

Type MS · Explain how villi aid absorption and the effect of damage

Answer 3 of 7 · Question 4 part b · 4 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how villi aid absorption and the effect of damage

Answer 4 of 7 · Question 6 part c.ii · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number					
6	(a)	(i)	18-12; 6;	accept 17.8/17.9	2
		(ii)	Two from: • volume of milk; • volume of the starter culture/live yoghurt; • type of starter culture • type of milk; • type of live yoghurt; • mixing;	reject 'amount'	2
		(iii)	• repeat the investigation at more temperatures; • between 20°C and 60°C;		2
		(iv)	• heated to kill (unwanted) bacteria in the milk; • cooled so that the bacteria in the starter culture are not killed;		2
	(b)		Any two from: • pH 4.4 inhibits the bacteria/bacterial growth; • denatures enzymes; • changes the shape of the active sites • lactose cannot bind to the enzyme/no enzyme-substrate complexes formed;		2
	(c)	(i)	Any two from: • the beads can be re-used; • product doesn't need to be treated to remove the enzyme; • large surface area to increase the rate of reaction; • enzyme more temperature stable;		2
		(ii)	• long/villi/microvilli to increase surface area; • concentration gradient maintained with blood flow; • short diffusion distance;		3

Total 15 marks

Type MS · Explain how villi aid absorption and the effect of damage

Answer 4 of 7 · Question 6 part c.ii · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how villi aid absorption and the effect of damage

Answer 5 of 7 · Question 7 part a · 3 marks

2023 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

7 (a)	Any three from • wall is one cell thick for faster diffusion/short diffusion distance; • close network of blood vessels to decrease diffusion distance/maintain gradient; • contains microvilli to increase surface area (for absorption/diffusion); • contains a lacteal/lymph vessels to absorb fat/fatty acids/glycerol;	Adaptation and explanation needed for each mark	3
(b) (i)	Any two from • (immune system) does not recognise self-antigens/sees self-antigens as foreign; • white blood cells/immune system produce antibodies; • antibodies bind to (self) antigens;		2
(ii)	• villi damaged/destroyed/less villi; • decreased surface area; • less/slower absorption of nutrients/named nutrient;		3

Total for Question 7 = 8 Marks

Type MS · Explain how villi aid absorption and the effect of damage

Answer 6 of 7 · Question 7 part b.ii · 3 marks

2023 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

Number				
7	(a)	Any three from • wall is one cell thick for faster diffusion/short diffusion distance; • close network of blood vessels to decrease diffusion distance/maintain gradient; • contains microvilli to increase surface area (for absorption/diffusion); • contains a lacteal/lymph vessels to absorb fat/fatty acids/glycerol;	Adaptation and explanation needed for each mark	3
	(b) (i)	Any two from • (immune system) does not recognise self-antigens/sees self-antigens as foreign; • white blood cells/immune system produce antibodies; • antibodies bind to (self) antigens;		2
	(ii)	• villi damaged/destroyed/less villi; • decreased surface area; • less/slower absorption of nutrients/named nutrient;		3

Total for Question 7 = 8 Marks

Type MS · Explain how villi aid absorption and the effect of damage

Answer 7 of 7 · Question 5 part c.i · 1 mark

2025 JUN · Paper 02 (regional R) · find part c.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.